FINAL YEAR PROJECT / SEMINAR

Federated learning for crop-disease detection

BANANA · CAULIFLOWER · GROUNDNUT · RADISH

AGRI VISION

Detecting crop symptoms. Comparing lightweight one-stage detectors.

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ABSTRACT

Agri Vision investigates crop-disease and pest-symptom detection using bounding boxes and disease labels. A three-client federated-learning simulation trains a shared detector across separate data partitions. The revised study covers banana, cauliflower, groundnut and radish across 24 classes. SSDLite, SSD–MobileNetV3 and EfficientDet-D0 are being compared with the retained historical YOLOv8n baseline.

PROBLEM STATEMENT

- Small lesions and similar symptoms make visual diagnosis difficult.
- Uneven class representation and variable image quality affect detection.
- A useful detector must balance precision, recall, localisation and computing cost.

AIM AND OBJECTIVES

Develop and assess an efficient federated crop-symptom detector.

- Use one verified 24-class dataset with fixed train, validation and test splits.
- Compare four complete one-stage detectors through separate experiments.
- Measure precision, recall, mAP, inference speed and model size.

METHODOLOGY

Shared 24-class dataset → fixed three-client partitions → independent detector experiments

CLIENT 1 • LOCAL TRAINING · **CLIENT 2 • LOCAL TRAINING** · **CLIENT 3 • LOCAL TRAINING**

TRAINING-IMAGE-WEIGHTED FEDAVG → GLOBAL DETECTOR

YOLOv8n · **SSDLite** · **SSD–MobileNetV3** · **EfficientDet-D0**

Train locally → aggregate → validate → save best/latest checkpoint → repeat

Each architecture has its own weights and run. Clients are simulated sequentially on one GPU; this is not a deployed multi-device network.

RESULTS AND CURRENT PROGRESS

PROVISIONAL 24-CLASS VALIDATION

Detector	Round	Precision	Recall	mAP50–95
SSDLite	10	66.44%	51.17%	39.06%
SSD–MobileNetV3	10	57.92%	47.61%	30.30%
EfficientDet-D0	2	56.75%	68.36%	46.68%

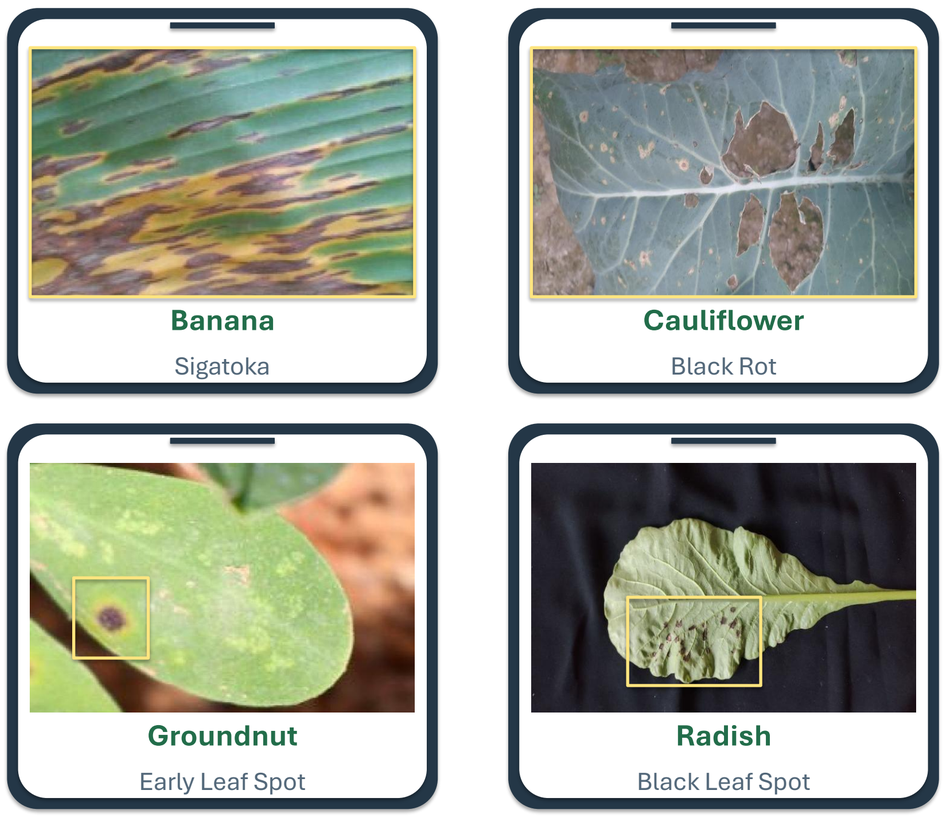
Same 4,157-image pooled validation split. P/R: confidence 0.25, IoU 0.50. EfficientDet has completed only two rounds; these scores do not establish a final winner.

HISTORICAL YOLOv8n • ROUND 40

Precision 66.27% Recall 57.43%

mAP50 59.55% mAP50–95 38.81%

30-class training; validation-image-weighted native client summaries. A common 24-class evaluation of this checkpoint is pending; the scores above are not directly comparable.



Labelled validation examples • yellow boxes are ground truth

TECHNOLOGIES USED

Python — Data & experiment tools

PyTorch — Detector implementations

Ultralytics — Historical YOLOv8n

Colab + Drive — GPU runs & checkpoints

SUMMARY

The revised study contains four crops and 24 classes. The historical round-40 YOLO baseline is retained. The other three detectors will continue toward round 30. Final conclusions require completed runs, common evaluation and measured deployment cost.

NEXT: complete training • evaluate retained YOLO on 24 classes • compare accuracy, speed and size

DATA: 14,514 train / 4,157 validation / 2,091 test • 3 simulated clients • Progress snapshot: 10 October 2026